ABSTRACT

Accurate demand forecasting plays a crucial role in improving operational efficiency, production planning, and supply chain management within export-oriented manufacturing industries. The natural fibre sector, characterized by fluctuating market demand, seasonal variations, and nonlinear sales behaviour, requires advanced forecasting techniques capable of capturing complex temporal patterns. This study proposes an intelligent demand forecasting framework that integrates statistical time-series modelling with machine learning approaches to enhance prediction accuracy for natural fibre products.

Historical sales data of three major product categories—Door Mat, Roll Mat, and Geo Textile Mat were analysed to identify underlying demand characteristics. Comprehensive data preprocessing techniques, including missing value handling, outlier treatment, exploratory data analysis, time-series transformation, and feature scaling, were applied to ensure data quality and modelling reliability. Stationarity analysis using the Augmented Dickey–Fuller (ADF) test revealed the presence of non-stationary behaviour, indicating strong trend and seasonal components within the sales data.

To evaluate forecasting performance, multiple models were implemented, including Autoregressive Integrated Moving Average (ARIMA), Seasonal ARIMA (SARIMA), Extreme Gradient Boosting (XGBoost), Hybrid ARIMA, and Hybrid SARIMA models. Model effectiveness was assessed using standard performance metrics such as Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and Mean Absolute Percentage Error (MAPE). Experimental results demonstrated that standalone statistical models struggled to capture nonlinear demand variations, while machine learning models alone lacked consistent temporal stability.

The proposed hybrid forecasting approach, combining seasonal time-series modelling with machine learning-based residual learning, significantly improved prediction accuracy across all product categories. Among the evaluated models, the Hybrid SARIMA framework achieved the lowest forecasting errors and highest reliability, confirming the effectiveness of hybrid modelling strategies for complex industrial datasets.

The outcomes of this research provide practical insights for demand-driven production planning, inventory optimization, and export management within the natural fibre industry. The study establishes that integrating statistical forecasting with machine learning techniques offers

a ii robust and scalable solution for real-world industrial demand prediction problems. Furthermore, the proposed framework contributes to the advancement of intelligent forecasting systems applicable to modern data-driven manufacturing environments.

Keywords: Demand Forecasting, Time Series Analysis, ARIMA, SARIMA, XGBoost, Hybrid Models, Machine Learning, Natural Fibre Industry, Sales Prediction.